DEWDRIVE · DESIGN DECISION RECORD

DewDrive TRL 2 review decisions

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/dewdrive		
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REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-02	Amish Chadha	O1 decided by Amish on 2026-10-02 (DWD-DEC-001, item 5)
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	Record Amish's decisions on the TRL 2 review items and the items that remain open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** accepted for items D1 to D8; the O2 proposals were decided in DWD-DDR-002; item O1 was decided by Amish on 2026-10-02 as recommended in the design decisions register (DWD-DEC-001, item 5)

Context

The TRL 2 review note (`docs/REVIEW.md`, session 2026-09-25, /populate) listed nine items as "Proposed, awaiting Amish". On 2026-09-25 Amish reviewed the review points for every portfolio repo and wrote: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them." Every item that carried a recommendation or a single proposed choice is therefore decided in favor of it. The one item without a recommendation stays open. TRL 4 work is on hold by the same instruction.

Options considered

The options for each item are those listed in `docs/REVIEW.md` (TRL 2 session) and in DWD-PRC-001 v0.2, Key design choices.

Decision

Table 1. Decided items.

#	Item	Decision
D1	Budget	Raise `budget_usd` from $400 to $500 for a research prototype with fan and logging, rather than a fanless $400 version or cost cuts that cost yield. Decided by Amish, 2026-09-25: go with recommendation.
D2	Problem wording	Reword the `project.yaml` problem line to: "Arid households without surface water or groundwater within reach depend on carried or trucked water; the air above them is an untapped supplementary source." Decided by Amish, 2026-09-25: go with recommendation.
D3	Sorbent	Silica gel with about 33 wt % $CaCl_2$, rather than LiCl composites, zeolite or MOF. Decided by Amish, 2026-09-25: go with recommendation.
D4	Collector layout	Single-stage flat glazed box with the condenser as its shaded floor, rather than dual-stage or a separate condenser. Decided by Amish, 2026-09-25: go with recommendation.
D5	Night airflow and power	Fan-assisted night airflow with a 10 W PV panel and a 12.8 V LiFePO4 battery, rather than natural airflow. Decided by Amish, 2026-09-25: go with recommendation.
D6	Flaps	Manual flaps, rather than automatic actuators. Decided by Amish, 2026-09-25: go with recommendation.
D7	Tilt and glazing	Fixed 20° tilt facing the equator; twin-wall polycarbonate glazing rather than glass. Decided by Amish, 2026-09-25: go with recommendation.
D8	Design point	20 °C and 40 % RH at night, 6.0 kWh/m² of sun per day on the aperture; low-humidity case 25 % RH. Decided by Amish, 2026-09-25: go with recommendation.

Notes on the decided items:

- **D1.** `budget_usd` in `project.yaml` is now 500. R11 in DWD-REQ-001 v0.3 is redefined to $500. The TRL 3 BOM totals $488 (DWD-CAL-001, J1).
- **D2.** The problem line is applied to `project.yaml` and `README.md`. The review recommended no change to the `pitch`, which still describes the device accurately, so it is kept.
- **D3.** The salt loading is decided at about 33 wt %. DWD-CAL-001 (section E) found that this loading lets the solution fill the pores above 51 % RH, and proposes about 25 wt % instead; that was a new item, decided by Amish on 2026-09-25 (DWD-DDR-002, D10: 25 wt %), which amends D3.
- **D4 and D5.** DWD-CAL-001 (section B) found that air flowing past the trays captures only about 11 % of the vapour it carries, so R1 and R2 are not met as drawn. Drawing the air through the bed was decided by Amish on 2026-09-25 (DWD-DDR-002, D9); it keeps D4 and D5.
- **D8.** DWD-CAL-001 (C1) shows that the 6.0 kWh/m² day puts 4.24 kWh/m², not 4.5, in the 09:00 to 15:00 window; DWD-REQ-001 v0.3 carries the corrected figure.
- **SwapCell.** DewDrive uses its own 76.8 Wh battery and does not use a SwapCell pack. The portfolio decisions of 2026-09-25 on the SwapCell interface (v0.3 items: wake without CAN, charge-while-discharging mode, latch vibration rating) and on pricing shared packs once therefore do not change the DewDrive design or BOM.

Table 2. Items left open by this record, both since decided.

#	Item	Status
O1	First partner and region for co-design, which will also supply the site climate data	**Decided by Amish, 2026-10-02:** first candidate type to approach (not yet agreed), a university dryland field station in a hot semi-arid region with night humidity of about 25 to 40 % and an existing weather record, with a water and sanitation NGO for the later community trial (DWD-DEC-001, item 5)
O2	New TRL 3 proposals: through-flow trays, salt loading, radiation between bed and condenser, humid-night protection	Decided by Amish, 2026-09-25: go with recommendation. Recorded in DWD-DDR-002 (D9 to D14)

Consequences

- DWD-PRB-001, DWD-PRC-001 and DWD-REQ-001 are revised to v0.3 to show the decisions: the key design choices of the precis are no longer "proposed", R11 carries the new budget, and the design point is decided.
- `project.yaml` carries `budget_usd: 500` and the reworded problem line.
- The partner question (O1) was decided on 2026-10-02 (DWD-DEC-001, item 5).